

Northstar Commerce Labs

These 12 self-contained labs build one coherent e-commerce and CRM application from contract design through a tested FastAPI/OpenAPI/SQLite release.

Lab	Build progression	Primary evidence
01	OpenAPI-first product/customer/order contract	Valid contract and prompt critique
02	First typed FastAPI routes	200 and structured 422 responses
03	Router/service separation	Stable OpenAPI route inventory
04	API-key dependency and CORS	Authorized and rejected requests
05	Product and CRM validation	Pydantic boundary failures
06	Output filtering and partial update	Stable response model and PATCH test
07	Error envelope and request trace	Response-to-log correlation
08	Regression suite and VS Code debugging	Failing-then-passing test evidence
09	Relational SQLite schema	Keys, checks and relationship proof
10	CRUD and atomic order transaction	Rollback and stock evidence
11	Responsive browser dashboard	Browser -> FastAPI -> SQLite trace
12	OpenAPI review and clean release	Test, start and secret-scan evidence

Every lab folder includes README.md/PDF, CHECKLIST.md/PDF, Prompts.md/PDF, ai_vibe.py, synthetic mock-data.json, a detailed Lab Guide DOCX/PDF and a complete working-files/ copy. The Python generation runner writes AI drafts only under working-files/generated/ for review.

The reusable demonstration project is in [demo-website](#).